

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2.	(a)	(i)		Callisto		1		1		
		(ii)		Moons of Jupiter get their heat from the Sun / Jupiter doesn't give out much heat / Jupiter isn't a star / Jupiter is a planet (1) They are [about] the same distance from the Sun / don't orbit the Sun (1)			2	2		
		(iii)		Callisto doesn't have the largest mean <u>diameter</u> [so claim not true] Alternative: Callisto has a smaller mean <u>diameter</u> than Ganymede [so claim not true]. Accept the converse. Accept correct reference to data.			1	1		
		(iv)		$671\,000 \times 2 = 1\,342\,000$ [km] [so not true] Alternative: $\frac{1\,070\,400}{2} = 535\,200$ km [so not true] Alternative: $\frac{1\,070\,400}{671\,000} = 1.6$ [so not true] Alternative: $1\,070\,400 - 671\,000 = 399\,400$ [so not true]			1	1	1	

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					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		Box 3. A light minute is the distance travelled by light in 60 seconds. Extra tick award 0 marks	1			1		
		(ii)		Mean distance = $\frac{588\,000\,000}{3.92}$ (1) = 150 000 000 (1) [km] accept 1.5×10^8 [km] or 150×10^6 [km]	1	1		2	2	
				Question 2 total	2	2	4	8	3	0